ANKUR VARMA

(551)-359-1942 | United States | ankur.varma@colorado.edu | linkedin/ankurvarma7 | ankurvarma7.github.io | github/ankurvarma7

EDUCATION

University of Colorado, Boulder Expected May 2026
Master of Science in Computer Science | ***GPA: 4.00/4.00*** *Boulder, CO*

- Lloyd Botway Fellow (Recognition of Outstanding Academics and Excellence in Research)

Indian Institute of Technology (IIT), Dhanbad May 2017
Bachelor of Technology | ***GPA: 3.96/4.00 (Rank 1, Gold Medal)*** *Dhanbad, India*

TECHNICAL SKILLS

Languages: Python, Java, C++, Rust, Julia, Scala | **AI/ML**: PyTorch, XGBoost, Naive Bayes, SHAP, LIME, vLLM, SGLang, Ray, scikit-learn, CNN, Transformer Models, RAG, spaCy, NLTK, HuggingFace, LangChain, LangGraph, Pydantic, CrewAI, MCP | **Cloud/DevOps**: AWS, GCP, Kubernetes, Docker, Terraform, Ansible, Linux, gRPC | **Frameworks**: Spring Boot, Fast API | **Databases**: Pinecone, PostgreSQL, MongoDB | **Big Data**: Kafka, Spark, Elasticsearch, Redis | **Tools**: Git, Jenkins, Cursor, Claude Code, Antigravity | **Concepts**: Microservices, REST, CI/CD, Agile, Cloud Computing

PROFESSIONAL EXPERIENCE

HighQube AI | ***Currently in stealth mode*** Dec 2025 – Present
Lead AI Systems Engineer | FastAPI, Docker, Gemini, asyncio, RapidFuzz, SQLAlchemy, PostgreSQL

- **Designed** a **6-stage AI/ML microservice** using **FastAPI** and **Docker** that automates extraction of complex ocean freight data from unstructured **250k+ token** Excel, PDF or document contracts, achieving **>98% F1 Score**.
- **Developed** an **LLM orchestration pattern** to bypass LLM output constraints; implemented a **multi-tier scouting system** that creates density-aware chunk plans, reducing end-to-end processing time by **65%**.
- **Implemented Gemini Context Caching** to maintain positional persistence across massive documents; optimized token throughput by embedding domain-specific logic into the cache layer, bypassing the "**multimodal tax**" of raw binary files.
- **Built** a **Scalable Asynchronous Ingestion Engine** using **asyncio** and **Semaphores**; enabled concurrent processing of document chunks with automated recursive splitting for high-density tables, increasing pipeline throughput by **500%**.
- **Architected** a **multi-turn autonomous Agentic AI framework** using **LangGraph** to automate complex document reconciliation.
- **Led Red Teaming and Adversarial Robustness** efforts to harden the pipeline against prompt injections and "**data-bombing**" attacks; developed a custom **Markdown Sanitizer** and integrated **NVIDIA NeMo Guardrails**.

University of Colorado Boulder May, 2025 – Dec, 2025
AI Research Assistant – LLM & Code Analysis | Pandas, Transformers, CodeBERT, AST Parsing, MOSS, Git, Jupyter *Boulder, CO*

- **Engineered** a **AI-detection pipeline** to assess academic integrity, analyzing **10,000+ student code submissions** against a corpus generated by **ChatGPT, Gemini, Claude, and industry tools (Copilot, Cursor)**, serving **28 course sessions**.
- **Developed** a **novel similarity engine** integrating **surface-level (Levenshtein)**, **statistical (code entropy, stylometry)**, **semantic (CodeBERT)**, and **structural analysis**. **Designed a custom AST matching algorithm** that normalized code to trees and used **recursive structural hashing** to quantify syntactic similarity. **Validated** the approach against Stanford's MOSS system, with scatter plot analysis confirming a **strong correlation (r=0.88)**, demonstrating the algorithm's **competitive accuracy**.
- **Built** an **automated data workflow** using **Python, Pandas, and regex** to parse **thousands of Jupyter notebooks**, extract code cells, and generate **differential comparisons (difflib)** for scalable analysis.

University of Colorado Boulder Jan 2025 – May 2025
Research Assistant – Probabilistic ML & Formal Methods | Julia, Turing.jl, MCMC, Bayesian ODE, Probabilistic Programming *Boulder, CO*

- **Pioneered** a **formal verification approach** by integrating probabilistic programming with **reachability analysis (flowpipe construction)**, generating the **first formally-bounded, high-confidence predictions** for the target infrastructure system.
- **Delivered state-of-the-art predictive accuracy**, achieving **100% coverage** of ground-truth observations within the **99% credible interval**, significantly outperforming standard posterior simulation baselines.
- **Co-authored** a **peer-reviewed conference paper** with Stanford University researchers, published in **Springer LNCS** (link).

Coal India Limited Nov 2017 – Jul 2024
Executive Engineer | SAP ERP, Python, SQL, Predictive Maintenance, ML Forecasting *Ranchi, India*

- **Drove data-informed procurement strategy** for **10,000+ SKUs**, integrating **SAP ERP** with **Python and SQL forecasting models** to optimize supply planning, achieving **over USD 1M in annual savings**.
- **Led** a **50+ person technical team** in process innovation and sustainable manufacturing, **boosting output by 90%** and generating $\sim$ **$250K in annual value** from scrap reutilization.

PROJECT

Adaptive Object Management in Ray
Proactive Placement and Eviction for AI Workloads | C++, Ray, Asynchronous Systems, Concurrency

- **Engineered the Enforcer component** by implementing **asynchronous watermark-driven spilling** to decouple memory reclamation from critical paths and eliminate synchronous spill stalls that freeze worker threads during memory-intensive phases.
- **Designed and implemented** the **background monitoring thread** using Ray's PeriodicalRunner API, proactively evaluating memory against configurable watermarks (80% high, 70% target) to maintain headroom without blocking worker execution.